

LUNA ARAFATUL NIKMAH

Medical Technology Innovator | R&D Project Leader | AI Diagnostics Researcher | Social Empowerment Driver
Surabaya, Indonesia.

+6285890008139 | [linkedin.luna](#) |  Portfolio_Luna.pdf

PROFESSIONAL SUMMARY

A highly driven Medical Technology undergraduate and R&D Project Lead specializing in AI-driven diagnostic systems and bio-signal processing. Demonstrated success in directing multidisciplinary teams to secure over \$43.498 in competitive grants, translating conceptual engineering into clinical trials and scalable hardware solutions.

SKILLS & TECHNOLOGIES

- **Medical & Life Sciences:** Applied Pathophysiology, Fundamental Biology for Diagnostics, Clinical Trial Logistics.
- **Project Management & Communication:** Cross-Functional Team Coordination, Grant Proposal Writing, Clinical Trial Logistics, Scientific Public Speaking & Pitching.
- **Machine Learning & AI (Applied):** Implementation of LSTM Networks & CNNs, Computer Vision, Predictive Analytics, Familiarity with AIoT Integration.
- **Programming & Data Tools:** Python, Data Processing for Bio-Signals (Acoustics & EEG Analytics).
- **Languages:** Indonesian (Native), English (Professional Working Proficiency).

EDUCATION

Institut Teknologi Sepuluh Nopember

Aug 2024-2028

Undergraduate **Bachelor of Engineering** Major in **Medical Technology**

GPA: 3.83/4.00

- Relevant Modules: Artificial Intelligence in Medicine, Medical Signal Processing (Acoustics & EEG), Medical Robotics, Machine Learning for Diagnostics, Medical Device Innovation, Bio-Sensors & Instrumentation, and Human-Centered & Safety System Design.

PROFESSIONAL EXPERIENCE (ENGINEERING & PROJECT MANAGEMENT)

Prognostic and Diagnostic Laboratory | R&D Project Lead - TB Vector

May 2025-Present

- Led the development and commercialization of TB-Vector (an AI acoustic beamforming and LSTM-based TB screening system), securing \$20,800+ in funding and Double Gold Medals at GEMASTIK.
- Orchestrated large-scale clinical trials with major hospitals (RSUA & RS Paru) and established an industrial supply chain for nationwide manufacturing and distribution (PT Xirka & MEDITRANS).

Tourism Empowerment Project, KPP MINING Youth in Action 2 | Core Team Member

Jan 2026-Present

- Secured \$1,280 (IDR 20M) from PT Kalimantan Prima Persada to co-implement an Asset-Based Community Development project in Desa Wisata Majapahitan Bejijong.
- Empowered over 760 local tourism stakeholders annually through targeted training in digital marketing, craft commercialization, and sustainable practices.

EEG-Based Neural Motion Lexicon Research (KATD ITS) | Project Leader

Dec 2025-Present

- Secured a \$5,900 Student Research Grant (SRG) to lead the development of a Brain-Computer Interface (BCI) for ALS patients within the ITS Artificial Intelligence and Digital Technology cluster.

REVELS, CancerFighters Team | Head of R&D for Breast Cancer Early Detection

Jan 2026-Present

- Direct research and development for REVELS commercialization, focusing on scaling medical hardware production and implementing *Microwave Vivaldi Antenna* to achieve precise 3D reconstructions of thermal anomalies for early-stage breast cancer detection.

Cilkankita Team (Innovillage PT. Telkom Indonesia) | AI System Integrator

Nov 2025-Feb 2026

- Directed an 11-person team to build an AIoT circular economy system, deploying AI and IoT sensors that empowered local workers to convert 5.4 tons of waste annually with a 3.82 SROI.

SafeStride Pertamina Foundation (PT Pertamina Indonesia) | Social Project Manager

April 2024-Dec 2025

- Led a multidisciplinary team to build and deploy SafeStride, a \$900 Pertamina Foundation-funded AI wearable, bridging engineering and healthcare to deliver direct monitoring and health education to 1,200+ geriatric users.

NUSAMED (BIMA Ministry of Education) | Head Research Assistant

Jan 2025-Dec 2025

- Led a student research team for the development of NUSAMED, an automated anamnesis and triage robot designed to streamline hospital administration and patient screening.
- Secured and managed a research grant of \$14900 (IDR 250M) from BIMA Ministry of Education, overseeing the project's strategic execution and financial accountability.

ORGANIZATIONS

ITS ABU Robocon Team

Nov 2024-Present

Official Manager

- Served as the Official Manager for the robotics team, leading financial management, sponsorship acquisitions, and media strategy. During this tenure, the team achieved prestigious accolades including 1st Place and the Best Strategy Award at the 2025 National KRAI, as well as the Best Design Award and the Kajima Award at the ABU Robocon 2025 in Ulaanbaatar, Mongolia.

SELECTED GRANTS & FELLOWSHIPS

- TB Vector (RIIM BRIN): AI-driven TB screening system commercialization (\$20,800).
- NUSAMED (BIMA Ministry of Education): Automated hospital anamnesis robot (\$14,900).
- Neural Motion Lexicon (SRG KATD ITS): EEG-based motor task recognition (\$5,900).
- CiIkan (Innovillage Telkom Indonesia): AIoT-based smart aquaculture system (\$1,300).
- Tourism Empowerment (KPP MINING): Community tourism resilience project (\$1,280).
- SafeStride (Pertamina Foundation): AI-driven fall detection wearable (\$900).

AWARDS

- Best Design Award & Kajima Award, ABU Robocon (Asia-Pacific Robot Contest), International (2025)
- Gold Medal (1st Place), Software Development, GEMASTIK, Ministry of Education and Culture, National (2025)
- Gold Medal (1st Place), Internet of Things (IoT), GEMASTIK, Ministry of Education and Culture, National (2025)
- 1st Place Winner, AI Ideation, National Research and Innovation Agency (BRIN), National (2025)
- 1st Place Winner, ABU Robot Competition Indonesia (KRAI), Ministry of Education and Culture, National (2025)
- Best Strategy Award, ABU Robot Competition Indonesia (KRAI), Ministry of Education and Culture, National (2025)
- 1st Place Winner & Best Writing, Electrolympic, UPN Veteran Jakarta, National (2025)
- 1st Place Winner, Policy Case Competition, NGO Nexus & Green Living, National (2025)
- 2nd Place Winner & Best Presentation, National Electrical Summit, Universitas Indonesia, National (2024)
- 3rd Place Winner, Electrolympic, UPN Veteran Jakarta, National (2024)
- 2nd Place Winner, Olimpiade Sains Nasional (OSN) Biology Kota Kediri (2023)
- 3rd Place Winner, ITS Biology Olympiade (2025)

INTELLECTUAL PROPERTY (PI)

- **Copyright (Computer Program): TB-Vector** – *Intelligent Software System for Spatial Tuberculosis Cough Tracking using Direction of Arrival (DoA) and Inverse Beamforming*. Registered under the Ministry of Law and Human Rights of Indonesia (ID: 000997340).
- **Copyright (Computer Program): CodeSight** – *Interactive Audio Gamification for Enhancing Algorithmic Thinking and Python Programming for Visually Impaired Students*. Registered under the Ministry of Law and Human Rights of Indonesia (ID: 001026489).
- **Copyright (Computer Program): Spatial Acoustic Surveillance for TB (SAS TB)** – *MEMS-based cough detection system for public space tuberculosis monitoring*. Registered under the Ministry of Law and Human Rights of Indonesia (ID: 000997344).

RECOGNITIONS

International Conferences & Leadership Engagements

- **Official Delegate, S-Tron Shanghai 2026**, in collaboration with BRIN and PT INTI Indonesia Makmur, Shanghai, China (2026).
- **Official Delegate, HPAIR Asia Conference 2026 (ANCOF)**, Harvard Project for Asian and International Relations (2026).
- **Official Delegate, One Young World ASEAN Leadership Forum 2026** (2026).
- **International Delegate, 2026 World Pre-Health Conference**, Harvard Undergraduate WPC via AMSA International (2026).

SPEAKERS

- **Invited Presenter** | Innovillage Community Service Social Project Showcase, Telkom University – National Level (2026)
- **Speaker** | GEMASTIK 2026 Hardskill Training, Univ. Padjadjaran – "Mastering Software Development: The Winner's Mindset and Method" (May 16, 2026)
- **Scientific Speaker** | University-Wide Freshmen Orientation, Institut Teknologi Sepuluh Nopember – University Level (2025)
- **Technical Trainer** | Official Internship Program, University Robotics Team (Robocon) – University Level (2025)
- **Speaker** | GENCO#2 Competition, SRE ITS & 180DC ITS – "Pitch. Compete. Level Up Your Business Game!" (May 3, 2026)
- **Speaker** | Achievement Workshop 2026, Dept. of Statistics, ITS – "Exploring GEMASTIK (National Student Exhibition in ICT) Competitions" (May 2, 2026)
- **Competition Judge** | Blue Camp, Department of Informatics, ITS – Department Level (2026)
- **Invited Speaker** | Faculty & Department Orientation (Electrical Engineering & Engineering Physics) – Faculty Level (2025)
- **Webinar Speaker** | Technical Sharing Session, University Robotics Team (Robocon) – University Level (2025)
- **Academic Mentor** | Medical Technology Scientific Session – Department Level (2025)
- **Public Speaker** | Student Activity Exhibition (Robotics Division) & Campus Community Forum – Community Level (2025)